The supplied third Janko families and their residue intersections

Private bounded computational proof

1 The literal population and its qualifications

The file `j3.rk3` in the fixed supplementary archive of L096 has 21 active rank-three families. Its final assignment interchanges construction entries 8 and 9; all active indices below refer to that final assignment. The literal permutations generate the same group G of order 50 232 960, identified as J_3 by the retained source model. Every supplied family passes actual flag-transitivity, firmness, residual-connectivity, residual-weak-primitivity, local-2-transitivity and rank-two-intersection tests. This is an exhaustive test of the supplied roster, not an enumeration of subgroups or a new classification.

The general all-residue theorem and the sparse right-coset shadow adapter are reused from their accepted packets. Primitivity is checked by the standard block–overgroup criterion written in the HS companion: a proper subgroup $H < G$ is maximal if and only if $\langle H, x\rangle = G$ for every outside (H, H)-double-coset representative. Degree one is excluded, negative cases retain actual intermediate subgroups, and each residue is checked until a primitive type is found or every type is exhausted. All 147 base residues, including panels, are checked on the literal 21 inputs. No earlier Janko census supplies a geometry predicate that this packet leaves untested.

L096 Tables 1 and 7 print a population of 17 for J_3 and counts $(5, 8, 4)$ for IP, WIP-only and failure of WIP. Our supplied-roster quotient has 19 classes and counts $(6, 8, 5)$. The discrepancy is retained explicitly. Global completeness of the earlier classification cited as L096 reference [19], D. Leemans, *Supplement to "The residually weakly primitive geometries of J_3"* (2003), is not reproved here, nor are its historical suffixes identified with active archive indices. Thus the number 19 below is exactly a supplied-roster conclusion.

2 A cheaper exact automorphism transporter

The full automorphism group computed from the literal group has order 100 465 920. Independently, the center of G is checked to be trivial, so $\mathrm{Inn}(G)$ has index two in $\mathrm{Aut}(G)$. The program selects and verifies an automorphism α that is not inner. Consequently $1, \alpha$ form a complete set of representatives modulo inner automorphisms. This reduction concerns the automorphism group; it does not require a splitting or an involutory choice of α.

For two ordered subgroup families, loop over all target type permutations and over $\beta \in \{1, \alpha\}$. Apply β to the whole input family. The remaining problem is simultaneous conjugacy by one element of G. Place target subgroups in decreasing order of size, with fixed tie order. Initially the permitted conjugating group is $C = G$. At each step find $x \in C$ conjugating the current entry to its target, conjugate the *whole* current family by x, and replace C by the normalizer of the target inside C. After the last placement no normalizer is needed. The product of the actual x's is recorded for positive matches.

This test is exact. Inner automorphisms are a normal subgroup of index two, so every automorphism has one of the two forms just tested. Once some entries have been placed, the possible

remaining conjugators form a coset of their common normalizer. If one transporter x to the next target is chosen, the other transporters differ from it by an element of that target normalizer. Induction therefore shows that each update preserves all possible solutions. Success gives an actual automorphism and one conjugator carrying the entire family to the target. Failure exhausts the two outer possibilities and every type permutation. The complete intersection-order profile is used only as a necessary rejection filter, as proved in the HS companion.

This exact permutation-group method completes all 21 rows, whereas a bounded direct action of the full automorphism group on subgroups stalled after seven rows. Those seven completed decisions are preserved and agree. A separate validation compares the new method with the direct full-automorphism method on 144 pairs of subgroup triples in A_5, including outer images and permuted targets; all 24 positive and 120 negative decisions agree. These finite validations support the implementation; the preceding argument proves the general reduction.

3 Explicit class separations and intersection results

For $J \subseteq \{1,2,3\}$ let $G_J = \bigcap_{j\in J} G_j$, with $G_\varnothing = G$. The multiset of all eight orders $|G_J|$ is invariant under group automorphisms and type permutations. On the supplied roster its buckets are the singleton entries and

$$\{7,8\}, \quad \{9,10,11\}, \quad \{13,14\}, \quad \{16,17\}.$$

There are 16 buckets. Actual complete-family transporters identify 13 with 14, and 16 with 17; their type permutations, automorphism images of every ambient generator, and the conjugating permutations are retained as exact data. The other repeated buckets split, with the following independent witnesses.

Entries 7 and 8 each have a unique parabolic of order 96; these parabolics contain respectively 27 and 15 involutions. Entry 9 has two parabolics of order 96, each with 27 involutions, whereas entries 10 and 11 have two such parabolics each with 15. These actual counts are obtained from complete conjugacy-class rosters in the small parabolic groups; group isomorphisms preserve them. Finally entry 10 satisfies full IP, while entry 11 fails WIP. The complete shadow certificates, with the failed directional intersections, therefore separate this last pair as well. These tests independently support the negative transporter decisions and give $16+1+2 = 19$ distinct supplied classes without relying on historical names.

Proposition 1. *Modulo entire-family automorphisms of the retained J_3 action and type permutations, the supplied roster has* 19 *eligible classes. Every class satisfies* IP_2*. Full IP, equivalently* IP_3 *here, holds in six classes; eight more satisfy WIP; five fail WIP.*

Proof. The order profiles separate different buckets. The explicit transporters identify exactly the two stated pairs, and the preceding invariants separate the other entries, so the quotient has 19 classes. All supplied entries were checked for the source geometry hypotheses, so these predicates transfer across the actual family isomorphisms.

For each representative, the accepted residue theorem enumerates all seven proper fixed-type subsets. Flag transitivity transports each base residue to all flags of those types. Panels are checked for geometry properties and only ranks two and three enter IP and WIP. This gives 133 base residues including panels, 76 of rank at least two, and 171 directional shadow decisions. Their actual all/exists quantifiers are reconstructed from the recorded values. The positive active representatives are

$$2, 3, 4, 7, 8, 10;$$

the weak-only representatives are

$$1, 5, 6, 12, 16, 18, 19, 21;$$

and the nonweak representatives are

$$9, 11, 13, 15, 20.$$

All rank-two residues satisfy IP, so the rank-three decision and full residual decision agree. This proves the asserted supplied-roster counts. □

4 Scope and reproducibility

The roster, complete programs, actual positive automorphisms, small subgroup conjugacy-class data, negative shadow witnesses and completed execution records are bound separately. GAP's exact finite group algorithms remain trusted. The default checker reparses the source, checks every residual quantifier and seventeen semantic controls, and preserves the 19 versus 17 discrepancy. Optional program replay recomputes the partition, exact witnesses, local invariants and all shadow decisions. The four original bounded eligibility batches are retained as complete executions and can be rerun separately; no large prior subgroup census is repeated.

No Lean formalization, complete historical-label correspondence, whole-original recovery or compression factor is claimed. The original Section 6 Magma listing and our complete programs remain visible for any later symmetric measurement boundary. The shared residue, maximality and transporter proofs are explicit dependencies intended for a single combined exposition, not unpaid imports from another topic.